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THROTTLING NAND READ-OUTS FOR IMPROVED HOST READ PERFORMANCE

Abstract

A method performed by a controller of a solid-state drive (SSD) having an integrated circuit comprising the controller. The method comprises receiving from a host a request for read data stored in a non-volatile semiconductor storage device of the SSD. The method also comprises determining if a capacity of internal memory of the integrated circuit is less than a predetermined fraction of its total capacity. Then the method comprises throttling a read-out rate used by the controller to retrieve data from the non-volatile semiconductor storage device if the capacity of internal memory of the integrated circuit is less than a predetermined fraction of its total capacity. The method then comprises retrieving the read data for storage in the internal memory using the throttled read-out rate.

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Background/Summary

FIELD

[0001] The present disclosure relates to solid-state drives (SSD) and methods for optimizing processing of read requests from a host so as to reduce read latency and improve performance.

BACKGROUND

[0002] A solid-state drive (SSD) generally has faster performance, is more compact, and is less sensitive to vibration or physical shock than a magnetic disk drive. Given these advantages, SSDs are being used in more and more computing devices and other consumer products in lieu of or in addition to magnetic disk drives, even though the cost-per-gigabyte storage capacity of SSDs is significantly higher than that of magnetic disk drives. SSDs utilize physical memory cells that comprise non-volatile semiconductor storage devices, such as NAND memory devices, to store data. A system-on-chip (SoC) controller of an SSD manages the transfer of data between a host and the memory cells of the SSD. Writing data to and reading data from the physical memory cells of an SSD typically involves transferring data between a host external to the SSD and the nonvolatile semiconductor storage devices.

[0003] SSDs often employ buffers to handle the transfer of data by the SoC when processing a read request or a write request received from a host. Such buffers include a larger amount of slower external memory and a smaller amount of faster internal memory for managing data transfers between the host and the NAND memory devices. However, the use of internal memory over external memory when servicing host requests has implications on the performance of the SSD. For example, the use of slower external memory for storing read data when processing a read request increases the read latency seen by the host. This has a knock-on effect on other processes that wait for the read data from the SSD.

[0004] In the case of processing read commands, the time in which read data is provided to the host is dependent on the data transfer rate of the data bus connecting the host to the SSD as well as the speed of the buffer used when reading out data from the NAND memory devices. Generally, the internal memory is preferred for use as the NAND buffer to ensure fast data transfers. However, the internal memory has a small capacity. Further, when the data read-out rate from the NAND memory device to the internal memory is higher than the data transfer rate of the bus between the host and the SSD, read data in the internal memory does not drain out to the host as quickly as it builds up within the internal memory. This leads to the data in the internal memory overflowing into the slower external memory. Draining read data in the external memory to the host is a slower process. This contributes to an increase in read latency of the SSD as seen by the host. The power consumption in processing a read request involving both internal and external memories would also increase. There is therefore a long felt need for optimized data transfers between a host and an SSD which reduce power consumption and reduce latency.

SUMMARY

[0005] According to an embodiment of the present disclosure, there is provided a method performed by a controller of a solid-state drive (SSD) having an integrated circuit comprising the controller. The method comprises receiving from a host a request for read data stored in a non-volatile semiconductor storage device of the SSD. The method also comprises determining if a capacity of internal memory of the integrated circuit is less than a predetermined fraction of its total capacity. Then the method comprises throttling a read-out rate used by the controller to retrieve data from the non-volatile semiconductor storage device if the capacity of internal memory of the integrated circuit is less than a predetermined fraction of its total capacity. The method then comprises retrieving the read data for storage in the internal memory using the throttled read-out rate.

[0006] According to another embodiment of the present disclosure, there is provided an SSD comprising a non-volatile semiconductor memory device comprising a plurality of memory dies for storing data. The SSD also comprises an integrated circuit comprising a controller and an internal

memory. The controller is configured to receive from a host a request for read data stored in a non-volatile semiconductor storage device of the SSD. The controller is also configured to determine if a capacity of internal memory of the integrated circuit is less than a predetermined fraction of its total capacity. The controller then throttles a read-out rate used by the controller to retrieve data from the non-volatile semiconductor storage device if the capacity of internal memory of the integrated circuit is less than a predetermined fraction of its total capacity. Next, the controller retrieves the read data for storage in the internal memory using the throttled read-out rate.

[0007] In certain implementations, throttling comprises reducing the read-out rate. In some implementations, the method comprises throttling the read-out rate when an available portion of the internal memory is less than a predetermined threshold. In other implementations, the method comprises providing the read data directly from the internal memory to the host. In certain implementations, the method comprises determining if the read data is stored in single or dual logical clusters in the non-volatile semiconductor storage device, and either (i) maintaining the read-out rate at a default value, or (ii) applying a reduced degree of throttling to a read-out rate, if the read data is stored in single or dual logical clusters.

[0008] In some implementations, the method comprises counting a number of logical storage units available in the internal memory, and generating a throttled read-out rate based on the count. In other implementations, the method comprises storing the throttled read-out rate and the count in a register within the controller. In certain implementations, the method comprises retrieving the read data for storage in the internal memory using a default read-out rate without throttling if the capacity of the internal memory is not less than a predetermined fraction of its total capacity. In some implementation, the throttled read-out rate used by the controller to retrieve read data from the non-volatile semiconductor storage device is less than the rate at which the read data is provided to the host. In other implementations, the request contains a logical address of the non-volatile semiconductor storage device in which the read data is stored.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] The foregoing and other objects and advantages will be apparent upon consideration of the following detailed description, taken in conjunction with the accompanying drawings, in which like reference characters refer to like parts throughout, and in which:

[0010] FIG. 1 illustrates a schematic representation of an SSD, configured according to one or more embodiments of the present disclosure;

[0011] FIG. 2 illustrates exemplary mapping of host read commands onto allocation units of an external DRAM buffer of the SSD in FIG. 1;

[0012] FIG. 3 illustrates an exemplary mapping of host read commands onto allocation units of an internal SRAM buffer as an alternative to or in addition to the external DRAM buffer in FIG. 2, according to one or more embodiments of the present disclosure;

[0013] FIG. 4A and FIG. 4B provide exemplary comparative performance data of the SSD in FIG. 1 with and without throttling the NAND read-out rate for various random and sequential read command sizes for single read queues, according to one or more embodiments of the present disclosure;

[0014] FIG. 5 provides exemplary comparative performance data of the SSD in FIG. 1 with and without throttling the NAND read-out rate for various random and sequential read command sizes for multiple read queues, according to one or more embodiments of the present disclosure;

[0015] FIG. 6 is a flow diagram of a method for processing a host read command by throttling the NAND read-out rate, according to one or more embodiments of the present disclosure; and

[0016] FIG. 7 is a flow diagram of a method for determining a degree of throttling to be applied

when processing the host read command of FIG. 6, according to one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

[0017] To provide an overall understanding of the devices described herein, certain illustrative embodiments will be described. Although the embodiments and features described herein are specifically described for use in connection with an SSD having a controller, it will be understood that all the components and other features outlined below may be combined with one another in any suitable manner and may be adapted and applied to other types of SSD memory architectures having a similar need to throttle read-out rates to improve latency of the SSD, among other benefits readily recognizable by person of ordinary skill in the art.

[0018] For a SSD with a hybrid memory system consisting of a larger amount of slower external memory and a smaller amount of faster internal memory, it is desirable for achieving good read performance that host read data should always reside within the internal memory. The present disclosure dynamically throttles the rate at which data is read-out from the NAND die based on the sampled fill state of the internal memory available for read-out data with the aim of minimizing the chance of any overflow into external memory, which would reduce performance. With a limited and relatively small amount of fast internal memory available for read data, increasing the amount of external memory available for read data can improve performance for some host traffic profiles. The increased amount of memory allows read commands to be started on more NAND die in parallel, while still maintaining a guarantee that memory will be available to accommodate the data at read-out time, thus increasing the rate at which read data becomes available from the NAND memory. For some traffic profiles the combinations of throughput and latency within the system can cause an overflow of read data into the slower external memory at read-out time if too much external memory is made available for processing read commands.

[0019] The amount of external memory that is configured to be available to read traffic cannot typically be varied as the host traffic profile varies and so may result in read-out data overflowing into the external memory for some profiles and causing poor performance. If overflow into external memory occurs, a large drop in performance is seen immediately as the read-out rate is limited by the total external memory bandwidth. Also, the buffer drain rate is reduced as the PCIe™ transfer rate is also limited by the total external memory bandwidth. NAND read-out and PCIe™ transfer to the host compete for the total external memory bandwidth and are thus both reduced compared to using internal memory.

[0020] By gradually reducing the NAND read-out rate as the internal memory approaches being full, the memory fill rate can be adjusted to closely match that value which can be supported by the memory drain rate via PCIe™ transfer of the read data to the host. Importantly the PCIe™ drain rate is not reduced due to having to read the data from external memory. An optimal performance for a wide range of read traffic profiles can thus be achieved. The throttling algorithm will ideally reduce the read-out rate by the smallest amount necessary to prevent overflow, while keeping the amount of unused internal memory available for reads at the smallest possible amount.

[0021] The present disclosure provides a method performed by a controller of a solid-state drive (SSD) having an integrated circuit comprising the controller. The method comprises receiving from a host a request for read data stored in a non-volatile semiconductor storage device of the SSD. The method also comprises determining if a capacity of internal memory of the integrated circuit is less than a predetermined fraction of its total capacity. Then the method comprises throttling a read-out rate used by the controller to retrieve data from the non-volatile semiconductor storage device if the capacity of internal memory of the integrated circuit is less than a predetermined fraction of its total capacity. The method then comprises retrieving the read data for storage in the internal memory using the throttled read-out rate. This drastically reduces read latency in the SSD, thereby improving the read performance of the device.

[0022] FIG. 1 is a block diagram of a memory system 100 comprising at least one host 110 in

communication with a storage device **120**. The host **110** is a computing system that comprises processors, memories, and other components as is generally known in the art, and which is not shown in FIG. **1** for the sake of brevity. Storage device **120** provides non-volatile storage and retrieval functionality for use by the host **110**. Storage device **120** may include an integrated circuit comprising a controller communicatively coupled to a memory array. Such an integrated circuit may comprise a system-on-chip (SoC) **130**. SoCs are advantageous as they provide a single integrated circuit that contains circuitry and components of the electronic system for the storage device **120** to function. SoC **130** may be communicatively coupled to a memory **140**. Memory **140** may comprise an array of NAND semiconductor memory, such as a NAND based flash memory device. The memory **140** may be organized into pages, blocks, clusters, planes, die and chips. Memory **140** may comprise a plurality of NAND chips, such as, for example, **32**, **64**, **128**, **256** separate NAND chips, and each NAND chip may be running separate read or write commands on individual die (not shown) within the chip. In the case of read commands, for example, the lack of coherence between read command queues accessing the memory **140** may cause varying amounts of collisions between read commands at the NAND die even when each read command queue maintains a consistent traffic profile.

[0023] Storage device **120** may include a memory external to the SoC **130**, such as a dynamic random access memory (“DRAM”) **150**. However, a person of ordinary skill in the art would understand that other types of external memory such as static random access memory (“SRAM”) or other suitable alternatives can be employed without departing from the scope of the present disclosure. External memory **150** comprises several buffers used to temporarily store data during read and write operations between the host **110** and the memory **140**. However, the external memory **150** is slower than the internal memory **170**, and thus use of the external memory **150** in read operations generally leads to a large reduction in performance.

[0024] Also shown in FIG. **1** is a memory controller **160** that enables the storage device **120** to perform various functions such as processing of read and write commands received from the host **110** as well as internal NVMe™ commands generated within the storage device **120** to maintain functionality. The storage device **120** may include a Read-Only Memory (ROM) that stores firmware for the operation of various states of the storage device **120**. The firmware comprises computer executable instructions that are executed by the controller **160** for operation of the storage device **120**. The storage device **120** also includes an internal memory **170**, such as a static random access memory (“SRAM”), for example, which forms part of the same integrated circuit as the SoC **130**. The internal memory **170** may be configured to store operational data such system metadata, mapping tables, NAND read-out rates and throttle thresholds, as well as data read from NAND memory clusters responsive to a host read request.

[0025] Storage device **120** may also comprise a front end host interface **132** which enables communication with the host **110** for the receipt of I/O commands such as read commands. Additionally, storage device **120** may include a back end memory interface **136** for communication with the memory **140** through a plurality of channels. Front end host interface **132** on the SoC **130** may comprise a Serial Advanced Technology Attachment (SATA) connector or a NVMe™ connector (NVMe™ is an acronym for “NVM express,” where “NVM” stands for “nonvolatile memory”) operating with a PCIe™ (“Peripheral Component Interface Express”) controller **133** and bus, for example. The PCIe™ controller **133** enables data transfer between host **110** and storage device **120** at a particular data transfer rate (e.g. 250 MB/s for PCIe™ 1.1, 500 MB/s for PCIe™ 2.0, and 1 GB/s for PCIe™ 3.0). These data transfer rates are specific to the generation of PCIe™ technology used and cannot be fixed. PCIe™ controller **133** facilitates the provision of data retrieved from NAND memory **140** to host **110** in response to a read request from host **110** at the above exemplified PCIe™ data transfer rates. Further, PCIe™ controller **133** facilitates storage of data from host **110** to NAND memory **140** in response to a write request from host **110**. Front end **132** may also include a NVMe™ buffer manager **134** that parses logical addressing information

contained in the host command for translation into physical address information for accessing the NAND memory **140**.

[0026] Back end memory interface circuitry **135** may comprise an Open NAND Flash Interface (ONFI) or a manufacturer's proprietary interface, for example. Back end circuitry **135** may comprise, for example, an interface according to, but not limited to: a Double Data Rate (DDR) memory bus standard such as DDR3, DDR4 or DDR5, a Low Power Double Data rate (LPDDR) memory bus standard such as LPDDR3, LPDDR4 or LPDDR5; a Hybrid Memory Cube (HMC) memory bus standard. Back end circuitry **135** includes a flash interface layer (FIL) **136** in communication with a NAND controller **137**. NAND controller **137** interacts with NAND memory **140** via data channels. In one implementation, FIL **136** communicates with a LUT engine **164** in SoC **130** to facilitate the processing of read and write commands received from host **110**. When processing read commands, for example, a logical block address (LBA) specified by the host **110** in a read command is used to index into a logical-to-physical look up table (LUT) in LUT engine **164**. The LUT engine **164** then provides to FIL **136** a physical block or cluster address in the NAND memory **140** from which data is to be read. FIL **136** provides this physical address to the NAND controller **137** which then issues low-level commands to NAND memory **140** for data read-out.

[0027] It should be understood that a 'read buffer' used to store data read-out from NAND memory **140**, as referred to herein, may comprise a combination of at least a portion of internal memory **170** and, optionally, at least a portion of external memory **150**. External memory **150** has a relatively small bandwidth, is larger in capacity, and slower at data transfer than internal memory **170**. With a limited and relatively small amount of fast internal memory **170** available for read data, performance can be improved for some host read traffic profiles by utilizing some external slower memory **150** of which there is usually a large amount that could be made available for data read out from the NAND memory **140**. The size of the portion of external memory **150** is selected such that an improvement in read throughput is seen by host **110**. The amount of external memory **150** will also vary depending on the read command size (or the size of read data requested by host **110**).

[0028] FIGS. 2 and 3 illustrate the structure of the read buffer with respect to external memory **150** and internal memory **170** when a read command is received from host **110**. In FIG. 2, the read buffer comprises external memory **150** only, while in FIG. 3, read buffer comprises external memory **150** and internal memory **170**. When the read command is received, front end **132** determines the size of the read data described by the read command and whether there is space in external memory **150** provisioned for the read buffer. If there is space in external memory **150**, controller **160** accepts the host read request and allocates read buffer space in the external memory **150** for the read data. Once read buffer space is allocated, front end **132** passes the LBA of each cluster within the read command to LUT engine **164**, which looks-up the address in the NAND memory **140** of the relevant cluster containing the read data and passes that address to FIL **136**. FIL **136** will make the decision as to when to schedule the read commands against NAND die in memory **140** based on the queue of cluster read requests that have accumulated from LUT engine **164**. This can be seen in the setup **200** of FIG. 2 at the time when the host read command is accepted by SSD **120** where the buffer ID of each host read command is associated with a start address of an allocation unit in external memory **150**.

[0029] When data read-out occurs from NAND memory **140**, controller **160** may determine that there is sufficient space in the faster internal memory **170** to hold the read data. Such just-in-time allocation is shown in the setup **300** of FIG. 3 where the buffer ID of certain read commands are associated with a start address of an allocation unit in internal memory **170** instead of the slower external memory **150**, and those allocation units in the internal memory **170** are actually filled the data read out from NAND memory **140**. This is especially the case for smaller data associated with read commands that involve single or dual cluster data (4 KB to 8 KB), for example. A content-addressable memory (CAM) is internal to memory controller **160** and used to decipher any overlay of host command buffer IDs with external memory allocation units and internal memory allocation

units. CAM is a type of memory that compares input search data against a table of stored data and returns address of matching data. In relation to FIG. 3, CAM receives as input a query with the host command buffer ID. If there is a hit, the CAM addresses the internal memory **170** for storage of data read out from NAND memory **140**. If there is no hit, the CAM addresses the external memory **150** for storage of data read out from NAND memory **140**.

[0030] However, the amount of external memory **150** available for read commands cannot generally be varied based on the traffic profile of the host **110** and thus a compromise between the selected size of external memory **150** and the corresponding drop in performance due to the slow speed of external memory **150** for a range of expected host traffic profiles must be carefully made. [0031] Variation in time of read throughput may be seen by the host **110** in the presence of multiple independent read command queues. This is because the lack of coherence between the read command queues may cause varying amounts of collisions between read commands at the NAND die in memory **140** even when each read command queue maintains a consistent traffic profile. To achieve good read performance during periods of more collisions a larger read buffer is useful as this allows NAND reads to be started on die that might otherwise be starved of commands. However, if such a larger read buffer is used during periods of fewer collisions, performance can be reduced due to the rate at which data is read-out from the NAND memory **140** to the read buffer being greater than the rate at which data can be drained from the read buffer via PCIe™ interface and returned to host **110**. This causes the read-data to overflow from internal memory **170** into slower external memory **150** which reduces performance of SSD **120**.

[0032] According to an embodiment of the present disclosure, this reduction in performance due to the use of external memory **150** to store read-data may be reduced or avoided by throttling (or reducing) the NAND read-out rate as the amount of available internal memory **170** approaches zero. Throttling may be triggered when the portion of available internal memory **170** falls below a predetermined start threshold. Throttling at the point of NAND read-out based on the fill level of internal memory **170** is advantageous as there is little in the way of feedback delay compared, for example, to sensing via an increase in visible command latency at host **110** and applying a throttle at the point of read command ingress from host **110** to SSD **120**. By attempting to keep as much read data as possible in internal memory **170**, the present disclosure reduces variations in read data throughput seen by host **110** as the host read traffic profile varies.

[0033] Further, the throttling start threshold and the reduction factor applied to the NAND read-out rate may be configurable depending on the size of read data involved when processing the respective read commands. For host traffic profiles with a mix of read command sizes and queue depths where there are some small randomly distributed host commands present within the mix, the throttling of the NAND read-out rate when the read is for only single or dual clusters of read data (4 KB to 8 KB) may increase the latency seen by host **110** for these small commands, without contributing much to avoiding overflow into external memory **150** as the bulk of the read data is due to the larger sequential traffic present in the mix. However, there is a potential issue if high rate random 4 KB to 8 KB host reads are the sole component of the host read traffic as this can cause overflow into external memory **150** for SSD device configurations that support multiple NAND die and fast NAND bus speeds. According to an implementation of the present disclosure, throttling of NAND read-out rate may exclude read traffic comprising single or dual cluster data where throttling is not triggered, or a small reduction factor in NAND read-out rate is applied, for small read data.

[0034] According to embodiments of the present disclosure, throttling of the read-out rate from NAND memory **140** to the NAND buffer begins to take effect once the available fraction of internal memory **170** drops below a threshold. Once below this threshold a linear reduction in read-out rate may be employed from the maximum read-out rate supported by the data channel between NAND memory **140** and SoC **130**. The controller **160** determines this reduction factor at the point where no more internal memory **170** is available for data read-out from the NAND memory **140**.

Other throttling techniques may be used to similarly maximize usage of internal memory **170** while preventing overflow of read data into the slower external memory **150**.

[0035] FIGS. **4A** and **4B** illustrate the effects of throttling the read-out rate from a NAND memory for host random read and host sequential read commands, respectively. The reduction in read-out rate has been modelled by adding a delay between each read-out from NAND memory **140**. As NAND read-outs can include anywhere from one to four 4K clusters of data, dependent on the host traffic profile, the delay has been modified by the count of clusters in the read-out to produce an average read-out rate per cluster. The average read-out rate per cluster is chosen to achieve a desired throttled read-out rate. As the modelled throttle takes effect on a direct measure of the remaining internal memory **170** for reads and acts directly on the next page read out from NAND memory **140**, the latency within the control loop is made as short as possible.

[0036] FIGS. **4A** and **4B** shows the modelled throughput of a single host reader with 8 MB of SRAM available for host read data as host command size varies. As can be seen, a 8 MiB (mega binary byte) read buffer provides sequential read performance that saturates a PCIe™ 6.0×4 bus for host command sizes of 8 KiB and greater, but the random read performance is significantly lower than this for intermediate-size host commands. Using a larger read buffer of 16 MiB improves the random read performance for some command sizes, but significantly reduces performance for others as read data overflows into external DRAM. The sequential read performance with this larger buffer causes overflow into external DRAM and a large drop in performance for all command sizes above 16 KiB. By using the aforementioned read-out throttling as described in the present disclosure, the larger 16 MiB read buffer can be used to boost random read performance for intermediate size read commands, while not reducing the performance for other random read command sizes. For sequential reads the performance with the larger 16 MiB read buffer with throttling remains unchanged from the throughput achieved with the smaller 8 MiB buffer size that matches the 8 MB size of the internal SRAM. The simulation results in FIGS. **4A** and **4B** illustrate the increase in throughput to the host once the read-out rate of the NAND memory is throttled as read command sizes increase. Also shown is the consistency of performance (30 GB/s throughput) of the SSD in processing read commands achieved with throttling the NAND read-out rate for read command sizes above 16 MiB.

[0037] FIG. **5** illustrates the modelled throughput of four concurrent uncorrelated sequential host read command queues. The same read traffic was presented to the model for three scenarios: unthrottled, slower NAND bus speed, and throttled NAND read-out rate. The total read data transferred is the same for all scenarios. Due to the different throughput during the periods of low and high collisions, the offsets along the time axis at which these periods occur are different on each plot. When there are fewer collisions between read command queues the NAND read-out rate can increase to the point that overflow occurs into the slower external memory resulting in lower performance, as can be seen by the low throughput of about 7.50 GB/s for the unthrottled scenario.

[0038] Statically restricting the NAND bus speed can prevent overflow when collisions are low, but also reduces performance for those periods where collisions are somewhat higher and the full NAND read-out bandwidth could otherwise be utilized. As seen from FIG. **5**, periods of higher performance in the unthrottled scenario have somewhat lower performance in the scenario with the slower bus (see throughput between 5 and 15 sec). The periods of low performance due to fewer collisions between host read queues causing overflow in the unthrottled scenario (see throughput between 25 and 45 sec) can be seen to have good performance in the slower bus scenario, which is also slightly better than the performance seen during the periods of higher collisions (see throughput between 5 and 15 sec) in the slower bus scenario. Throttling the NAND read-out rate as detailed in the aforementioned description avoids overflow of the internal SRAM and also maximizes performance when collisions are high enough that overflow cannot occur (see throughput between 5 and 15 sec for throttled scenario).

[0039] FIG. **6** illustrates an exemplary flow diagram of a method **600** of processing a read

command received by the SSD **120** from a host **110**, according to an embodiment of the present disclosure. Method **600** relates to an improvement on the throughput of the SSD **120** in processing read commands by throttling the read-out rate of data from the NAND memory **140** such that the read data does not overflow from the internal SRAM **170** to the slower external DRAM **150**, as described above.

[0040] Method **600** begins at step **610** where front end **132** waits for a read command from host **110**. Once a read command is received, controller **160** determines in step **620** if a read buffer is available for the read data clusters as identified in the read command. If the read buffer is unavailable to hold the read data clusters ('N' at step **620**), controller **160** waits. If the read buffer is available to hold the read data clusters ('Y' at step **620**), front end **132** issues lookup commands to LUT engine **164** for each cluster of read commands (step **630**). Next, in step **640**, LUT engine **164** forwards the cluster lookup results to FIL **136** in back end **135**. FIL **136** then forwards the NAND read commands for each cluster to NAND controller **137** (step **650**). Here NAND controller **137** issues low-level commands to NAND memory **140** for an array-read (step **655**). FIL **136** then waits for data to be ready for read-out from the NAND memory **140** (step **660**). NAND controller **137** polls the NAND memory **140** for an array-read complete signal in step **665**.

[0041] Next, FIL **136** requests memory controller **160** to overlay allocation units in the DRAM (external memory **150**) with allocation units in the SRAM (internal memory **170**), per step **670**. The NAND controller **137** then issues low-level read-out commands to NAND memory **140** (step **675**). FIL **136** then performs a read-out of clusters into memory allocation units using a stored throttled read-out rate (step **680**). The controller also runs a read-out throttle iteration in step **685**, involving throttling of the read-out rate (step **690**), described in FIG. 7. Next, in step **695**, the memory allocation units in the DRAM and possible overlaid SRAM are released.

[0042] A read-out throttle iteration may occur when the available capacity of internal memory **170** falls below a predetermined fraction of its total capacity. The throttle iteration begins in step **710**. Here the controller performs a sample count of the available SRAM allocation units in step **720**. A throttling algorithm is then applied to the new available count in step **730**, where internal parameters of the throttling algorithm are updated. Next in step **740** a throttled cluster read-out rate is generated from the internal parameters. The throttle read out rate is then stored (step **750**) for use in the next iteration of the process **600** beginning at step **610**.

[0043] As seen in the foregoing, by throttling the NAND read-out rate as the capacity of internal memory **170** falls below a predetermined fraction of its total capacity, the internal memory fill rate can be adjusted to closely match that value which can be supported by the memory drain rate via PCIe™ transfer of the read data to the host. This minimizes the chance of any overflow into external memory, and thus improves performance.

[0044] In the foregoing, all recitation of "manager" and "engine" should be taken to mean a plurality of circuits within the controller that facilitates the function as described. Such circuits may comprise electronic components formed on a semiconductor chip, such as, for example, transistors and resistors. It should be noted that the term "about" or "approximately" in the foregoing indicates a range of $\pm 20\%$ of the stated value. Additionally, in the foregoing, all recitation of "command," "action," "operation" or "function" should be taken to be based on algorithms and instructions stored on a non-transitory computer-readable medium, that, when executed by a processor, causes a controller of an integrated circuit of a solid-stated drive (SSD) to perform the command, action, operation or function. All recitation of "device," "memory," "array" and "dies" are used interchangeably when used in relation to the NAND non-volatile semiconductor memory device. The term "similar" as used herein indicates close to identical but for a stated difference.

[0045] Other objects, advantages and embodiments of the various aspects of the present invention will be apparent to those who are skilled in the field of the invention and are within the scope of the description and the accompanying drawings. For example, but without limitation, structural or functional elements might be rearranged consistent with the present invention. Similarly, principles

according to the present invention could be applied to other examples, which, even if not specifically described here in detail, would nevertheless be within the scope of the present invention.

Claims

1. A method performed by a controller of a solid-state drive (SSD) having an integrated circuit comprising the controller, the method comprising: receiving from a host a request for read data stored in a non-volatile semiconductor storage device of the SSD; determining if an available capacity of internal memory of the integrated circuit is less than a predetermined fraction of its total capacity; throttling a read-out rate used by the controller to retrieve data from the non-volatile semiconductor storage device if the available capacity of internal memory of the integrated circuit is less than the predetermined fraction of its total capacity; determining a size of the read data associated with the request; and retrieving the read data for storage in the internal memory using the throttled read-out rate only if the available capacity of internal memory of the integrated circuit is less than the predetermined fraction of its total capacity and the size of the read data is above a specified size.
2. The method of claim 1, wherein throttling comprises reducing the read-out rate.
3. The method of claim 1, further comprising: throttling the read-out rate when an available portion of the internal memory is less than a predetermined threshold.
4. The method of claim 1, further comprising: providing the read data directly from the internal memory to the host.
5. The method of claim 1, wherein the specified size corresponds to single or dual logical clusters in the non-volatile semiconductor storage device; and wherein if the size of the read data is at or below the specified size, retrieving the read data is performed at either (i) a default value of the read-out rate, or (ii) a read-out rate that is faster than the throttled read-out rate.
6. The method of claim 1, further comprising: counting a number of logical storage units available in the internal memory; and generating a throttled read-out rate based on the count.
7. The method of claim 6, further comprising: storing the throttled read-out rate and the count within the controller for a subsequent read iteration.
8. The method of claim 1, further comprising: retrieving the read data for storage in the internal memory using a default read-out rate without throttling if the capacity of the internal memory is not less than a predetermined fraction of its total capacity.
9. The method of claim 1, wherein the throttled read-out rate used by the controller to retrieve the read data from the non-volatile semiconductor storage device is less than the rate at which the read data is provided to the host.
10. The method of claim 1, wherein the request contains a logical address of the non-volatile semiconductor storage device in which the read data is stored.
11. A solid-state drive (SSD) comprising: a non-volatile semiconductor memory device comprising a plurality of memory dies for storing data; and an integrated circuit comprising a controller and an internal memory, the controller configured to: receive from a host a request for read data stored in a non-volatile semiconductor storage device of the SSD; determine if an available capacity of internal memory of the integrated circuit is less than a predetermined fraction of its total capacity; throttle a read-out rate used by the controller to retrieve data from the non-volatile semiconductor storage device if the available capacity of internal memory of the integrated circuit is less than the predetermined fraction of its total capacity; determine a size of the read data associated with the request; and retrieve the read data for storage in the internal memory using the throttled read-out rate only if the available capacity of internal memory of the integrated circuit is less than the predetermined fraction of its total capacity and the size of the read data is above a specified size.
12. The SSD of claim 11, wherein throttling comprises reducing the read-out rate.

- 13.** The SSD of claim 11, wherein the controller is further configured to: throttle the read-out rate when an available portion of the internal memory is less than a predetermined threshold.
- 14.** The SSD of claim 11, wherein the controller is further configured to: provide the read data directly from the internal memory to the host.
- 15.** The SSD of claim 11, wherein the specified size corresponds to single or dual logical clusters in the non-volatile semiconductor storage device; and wherein if the size of the read data is at or below the specified size, the controller is further configured to retrieve the read data is performed at either (i) a default value of the read-out rate, or (ii) a read-out rate that is faster than the throttled read-out rate.
- 16.** The SSD of claim 11, wherein the controller is further configured to: count a number of logical storage units available in the internal memory; and generate a throttled read-out rate based on the count.
- 17.** The SSD of claim 16, wherein the controller is further configured to: store the throttled read-out rate and the count within the controller for a subsequent read iteration.
- 18.** The SSD of claim 11, wherein the controller is further configured to: retrieve the read data for storage in the internal memory using a default read-out rate without throttling if the capacity of internal memory of the integrated circuit is not less than a predetermined fraction of its total capacity.
- 19.** The SSD of claim 11, wherein the request contains a logical address of the non-volatile semiconductor storage device in which the read data is stored.
- 20.** The SSD of claim 11, wherein the throttled read-out rate used by the controller to retrieve the read data from the non-volatile semiconductor storage device is less than the rate at which the read data is provided to the host.
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